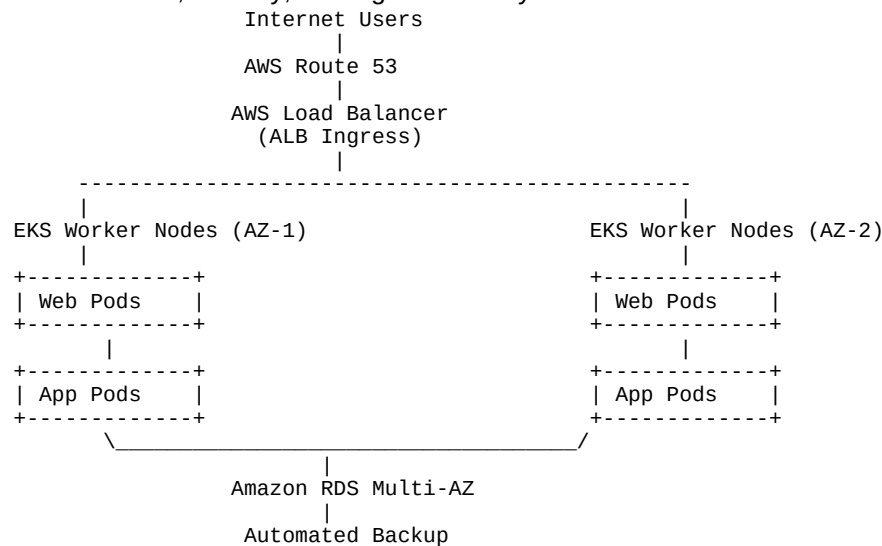


Designing a Highly Available Multi-Tier Application on AWS EKS

Scenario: Deploy a highly available multi-tier application in AWS using Amazon EKS. The application consists of a Web Tier, Application Tier, and Database Tier. The design must ensure scalability, fault tolerance, security, and high availability.



Architecture Design

1. Amazon EKS Cluster

Deploy the EKS cluster across multiple Availability Zones (AZs) to eliminate a single point of failure. Worker nodes should be distributed across at least two AZs using managed node groups.

2. Web Tier

Deploy frontend containers as Kubernetes Deployments with multiple replicas. Use Horizontal Pod Autoscaler (HPA) to automatically scale pods based on CPU or memory usage.

3. Application Tier

Backend services run in separate Kubernetes Deployments. Use ReplicaSets and multiple replicas to ensure application availability during failures.

4. Load Balancing

Use AWS Application Load Balancer (ALB) with Kubernetes Ingress Controller. The ALB distributes incoming traffic across healthy pods running in different Availability Zones.

5. Database Tier

Use Amazon RDS Multi-AZ deployment for automatic failover and high availability. Enable automated backups and read replicas if required for heavy read workloads.

6. Storage

Use Amazon EBS CSI Driver for persistent volumes. For shared storage requirements, use Amazon EFS.

7. Security

Implement IAM Roles for Service Accounts (IRSA), Security Groups, Network Policies, AWS Secrets Manager, and private subnets for backend components.

8. Monitoring and Logging

Use Amazon CloudWatch, Prometheus, and Grafana for monitoring. Use Fluent Bit to send container logs to CloudWatch.

Conclusion

A highly available EKS solution should distribute workloads across multiple AZs, use autoscaling, load balancing, and RDS Multi-AZ databases. This design provides fault tolerance, scalability, security, and operational reliability suitable for production environments.